Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in M≥6.5 Earthquake Catalogs, 1973–2026 CE

Temporal ETAS null and hold-out validation

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Abstract.

We analyze an ***analysis catalog of 4,267 unique M≥6.5 events*** *(modern window 1973–2026: 2,041). Baiesi–Paczuski η detector yields* ***27 algorithmic candidates****.* ***Catalog-calibrated temporal ETAS: p_ETAS=1.0*** *(in-sample null; full numbers — Sec. 4.1). Spatial component not modeled. Pre-1900 NOAA — Supplementary S3.*

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi–Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that M≥7 clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global M≥7 and M≥8 rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, **falsify**) the hypothesis that physically meaningful multi-regional global series exist, with **primary inference on the modern window (1973–2026)**, using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. **Scope.** We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different η-linkage statistic but does not supersede their conclusions.

§1.1 Research question. Do physically meaningful multi-regional global series exist in M≥6.5 (1973–2026)? (a) Permutation → see Sec. 4.1. (b) ETAS MLE → see Sec. 4.1. (c) Global-series hypothesis: **not tested** by temporal-only ETAS. (d) WLS (Supplementary S2): coupling illustration, not primary null.

2. DATA AND METHODS

2.1 Catalog compilation

Magnitude notation. Catalog thresholds and detector gates use **M≥6.5** (USGS ComCat, predominantly M_w); individual events are cited with catalog type where relevant (e.g. M_w 7.3).

Three catalogs were merged: **USGS ComCat** (1900–2026, primary instrumental); **ISC Bulletin** (relocated hypocenters); and **NOAA NGDC** (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using ±30 days and ≤50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains **4,267** unique M≥6.5 events (from **4,418** CSV rows; ~151 M<6.5 excluded from clustering). USGS ComCat: **2,088** raw (1973–2026) → **2,041** after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). quality_score is metadata, not an inclusion filter. **Primary analysis set:** events 1900–2026 (4,218); 47

pre-1900 records remain in CSV (provenance) but are excluded from the primary detector pipeline and ETAS calibration window (1973–2026 only) — see Supplementary S3. **Final catalog:** 4,267 M≥6.5 events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields M_c = 6.55. Maximum-likelihood b-value from 1,688 events above M_c:

b = 0.911 +/- 0.018 (n = 1,688 events)

2.2 Tectonic heuristic (deprecated)

The Bird (2003) tectonic-path heuristic is **excluded from the primary pipeline**; **no synthetic benchmark** in this work; comparison deferred to supplementary material and future work. Primary analysis uses **great-circle distance only** (see paper/supplementary.md §S1).

eta_ij = t_ij * r_ij^1.6 * 10^(-b*mi)

*b=1.0 — deliberate Baiesi & Paczuski (2004) simplification; catalog b=0.911±0.018 for M_c/completeness and MC null only — **not** in the η formula. Full pipeline at b=0.911 (run_sensitivity_b0911_full.py): N=27, Jaccard=1.0; **8.2%** upstream cluster-label mismatch — detector gates dominate. Changing b re-runs upstream identify_clusters() labels only, not global_series() gates (sensitivity_b0911_full_pipeline.json).*

2.3 Analysis pipeline

Raw catalogs → dedup → exclude M<6.5 (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: η NN + KDE threshold). At global M≥6.5 scale ZBZ is permissive (sparse events → high η). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, N=27 likely unchanged (untested).

Why GK–ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low η at global catalog scale. Under fixed gates GK/ZBZ/none all yield N=27 (sensitivity_declustering.json); different mainshock labels (24 vs 1) could matter for cluster analysis.

Detection algorithm (§3.3). GK WINDOWS, T(M)/R(M) interpolation, magnitude-descending; aftershocks [0,T], foreshocks [−T/2,0); haversine. find_nearest_neighbor: causal argmin η, b=1.0, GC km. identify_clusters: Union–Find, η_0 KDE. global_series: used[] mask, window [t_i, t_i+Δt), mean GC>1500 km; merge 142→47. **Gates:** N≥4; M≥6.5; mean pairwise GC>1500 km. Flinn–Engdahl zone count

— diagnostic only.

2.4 Primary ETAS null

Primary ETAS null uses catalog-calibrated temporal MLE on GK mainshocks (calibrate_etas_mle.py). Hold-out: train 1973–2000 (1024 GK mainshocks), hold-out 2001–2026 (1010 events); MLE **train only**, parameters fixed without retraining (calibrate_etas_holdout.py; N_obs=13, p=1.0). WLS — Supplementary S2 only.

2.5 Statistical validation

Permutation statement: p=0.0001 rejects **Poisson event times only** (Ogata 1988); does not confirm global series. ETAS — Sec. 4.1. BH post-hoc — not discovery.

3. RESULTS

3.1 Identified series

Full historical analysis yields **47 detector candidates** (not validated global series): 27 modern (p ≤ 0.0001), 15 early instrumental (p = 0.007; pre-1960 incompleteness caveat), 5 historical candidates (**not significant**, p = 0.46; 47 M≥6.5 events pre-1900). 142 cluster candidates before filtering. *Interpretation — Sec. 4.*

Series ID	Events	Regions	Mmax	Years	Mean lat°	Mean lon°	Notes
1905–1910	193	43	8.8	1905–1910	−60…72	−180…180	Early instrumental; incomplete before ~1960
S047	53	5	8.0	1982–2024	−21.5…18.9	−175.5…155.2	Western Pacific
S170	46	12	9.1	2002–2023	−14.3…33.8	−113.1…167.3	Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	−8.1…16.5	120.4…146.4	Western Pacific arc
S116	22	5	8.2	1993–2021	−31.7…10.8	−179.5…179.4	South Pacific

Table 1. Raw detector candidates — NOT validated series; illustrative only.

3.2 ETAS validation (Sec. 4.1 canonical)

Split	N_obs	mean	p_ETAS
In-sample (1973–2026)	27	27.0	1.0
Train-calibrated hold-out (2001–2026)	13	13.0	1.0

Permutation: p = 0.0001 (1/10,001), z = -6.17. Hold-out — partial out-of-time check, not spatial validation. **Multiple testing:** the 27 modern series were **not** FDR-corrected for the 142-window search; Bonferroni α/142 ≈ 0.00035 > p = 0.0001. BH post-hoc on N=47 — not a discovery claim.

3.3 Spatial–temporal distribution

Elevated detector candidate frequency (not validated physical series) occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median $\Delta\log_{10}\eta = +0.28$ on random pairs (mostly 1.5× GC fallback).

3.4 Consolidated sensitivity table (modern window)

Parameter	Setting	N_series
GC gate	1000 km	27
GC gate	1500 km (baseline)	27
GC gate	2000 km	27
Window	1 yr	53
Window	2 yr (baseline)	27
Window	5 yr	11
Window	10 yr	6
b in η	1.0 (BP 2004)	27
b in η	0.911 (catalog)	27
b overlap (full pipeline)	Jaccard=1.0; upstream 8.2%‡	27
Declustering	GK / ZBZ / none	27 / 27 / 27
min_events (strict)	5 / 6 / 8	27 / 27 / 27
Catalog	GK mainshocks only	27
η_0 ±20%	not_applied	—

*Table 2. N_series sensitivity under fixed detector gates (mean GC > 1500 km, N ≥ 4). Sources: sensitivity_*.json. ‡**8.2%** = 165/2017 GK mainshocks whose identify_clusters() cluster label changes when b goes from 1.0 to 0.911 (full pipeline; sensitivity_b0911_full_pipeline.json).*

The row **b overlap (full pipeline)** reports a full re-run GK → *identify_clusters()* → *global_series()* at b = 1.0 vs 0.911. **Jaccard = 1.0** means identical **series event sets** across 27 detections --- *global_series()* does not use b in η. **8.2%** upstream is the fraction of GK mainshocks whose **η-cluster label** changes at *identify_clusters()*; this is a different level from downstream series membership, so N = 27 and Jaccard = 1.0 do **not** prove unchanged upstream structure.

N = 27 stability. At fixed gates the same event kernel survives merge 142→47→27 under GK/ZBZ/none (Jaccard of event sets = 1.0; series membership Jaccard = 0.32 for ZBZ/none). Window width dominates (53 at 1 yr, 11 at 5 yr). Detector-artifact stability, not a physical invariant “core 27.”

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4. DISCUSSION AND CONCLUSIONS

Temporal ETAS: No excess candidates beyond catalog-calibrated null (Sec. 3.2). Spatial linkage not tested. Permutation rejects Poisson times only. Bird/WLS/pre-1900 — paper/supplementary.md §S1–S3.

Supplementary material. Bird (2003), WLS negative control, pre-1900 NOAA: paper/supplementary.md (§S1–S3).

Limitation	Affected step	Impact on main conclusion
η_0 unverified at global scale	GK/ZBZ identify_clusters	KDE valley ≈ 7.1×10^{-6}; global_series skips η_0 → N=27; mis-specified η_0 shifts GK/ZBZ labels only
b = 1.0 vs 0.911 in η	η upstream	N_series = 27 both; 8.2% upstream label mismatch (full pipeline)
No spatial Ogata MLE	ETAS null	Temporal MLE primary; spatial MLE future work
142 windows + merge	Detector	Main source of liberalness
Calibrated p_ETAS = 1.0	ETAS test	Consistent with null; supports negative conclusion

Table 3. Limitation → affected step → impact on main conclusion (Sec. 5.6).

Impact analysis. N = 27 is computed by global_series() without η_0 filtering; full pipeline b = 1.0 vs 0.911 leaves N_series = 27, Jaccard = 1.0, but 8.2% upstream label mismatch. 142 sliding windows dominate detector liberalness.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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